PREPARE-THRIVE-SUCCEED

IPU NOTES

B.TECH CSE

2nd Year

SNAPED

COMMUNITY

CM (Unit-3)

to read today

JOIN US:

Existence of Solution

Systems of linear non-homogeneous equations:

Let

$$\left.\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \cdots\cdots\cdots\cdots &\cdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{m3}x_3 &= b_m \end{aligned}\right\} \quad (I)$$

be a system of m non-homogeneous equations in n unknowns $x_1, x_2, \cdots, x_n$:

If $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & \cdots & a_{mn} \end{bmatrix}_{m\times n}$, $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}$, $B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1}$

then the system (I) can be written as

$$AX = B \quad (2)$$

Any set of values of $x_1, x_2, \cdots, x_n$ which simultaneously satisfy all these equations is called a solution of the system (I). When the system of equations has one or more solutions, the equations are said to be consistent, otherwise they are said to be inconsistent.

$$A^* = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

is called the augmented matrix of the given system of equations.

Th. 1. The system (I) of non-homogeneous linear equations has a solution iff the coefficient matrix A and the augmented matrix A* have the same rank.

Th. 2. The system (I) has a unique soln. iff Rank of A = m = n.

Th. 3. If $r<n$ the system (I) has infinite solns out of which it will have $n-r+1$ linearly independent solutions where r is the rank of the coefficient matrix. In this case r of the unknown variables can be expressed in form of the remaining $n-r$ variables which can be assigned any arbitrarily chosen values.

Important: While finding the rank, we use any row operation i.e. using the echelon form.

Ex. 1 Show that the equation

$$x-4y+7z=14$$
$$3x+8y-2z=13$$
$$7x-8y+26z=5$$

are inconsistent.

Soln.
$$\begin{bmatrix} 1 & -4 & 7 & 14 \\ 3 & 8 & -2 & 13 \\ 7 & -8 & 26 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & -4 & 7 & 14 \\ 0 & 20 & -23 & -29 \\ 0 & 20 & -23 & -93 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -4 & 7 & 14 \\ 0 & 20 & -23 & -29 \\ 0 & 0 & 0 & -64 \end{bmatrix}$$

which is in echelon form with three nonzero rows. $\therefore \rho(A^*)=3$. Also $\rho(A)=2$.

$\therefore$ The system is inconsistent.

(2) Solve the following system of equations by matrix method.

$x+y+z=6$, $x-y+2z=5$, $3x+y+z=8$

Soln:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 3 & 1 & 1 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 6 \\ 5 \\ 8 \end{bmatrix}$$

$\therefore$ Given system is $AX = B$.

Augmented matrix $A^* = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & -1 & 2 & 5 \\ 3 & 1 & 1 & 8 \end{bmatrix}$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & -2 & 1 & -1 \\ 0 & -2 & -2 & -10 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & -2 & 1 & -1 \\ 0 & 0 & -3 & -9 \end{bmatrix}$$

which is echelon form with three non zero rows.

$\therefore \rho(A^*) = 3$,

Also $\rho(A) = 3$

$\therefore \rho(A) = \rho(A^*) = 3 =$ No. of unknowns.

$\therefore$ System has ~~infinite~~ unique soln.

Now the system is reduced to

$$\left.\begin{aligned} x+y+z &= 6 \\ -2y+z &= -1 \\ -3z &= -9 \end{aligned}\right\} \Rightarrow$$

$z=3$, $y=2$ & $x=1$. by back substitution.

$\therefore \boxed{x=1, \; y=2, \; z=3}$

③ Solve

$2x+4y-z=9, \quad 3x-y+5z=5, \quad 8x+2y+9z=19$

<u>Soln:</u> The given system can be written as

$$\begin{bmatrix} 2 & 4 & -1 \\ 3 & -1 & 5 \\ 8 & 2 & 9 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 5 \\ 19 \end{bmatrix}$$

$$A^* = \begin{bmatrix} 2 & 4 & -1 & 9 \\ 3 & -1 & 5 & 5 \\ 8 & 2 & 9 & 19 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 & -1 & 9 \\ 0 & -7 & 13/2 & -\frac{17}{2} \\ 0 & -14 & 13 & -17 \end{bmatrix}$$

$$\sim \begin{bmatrix} 2 & 4 & -1 & 9 \\ 0 & -7 & 13/2 & -\frac{17}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

which is in echelon form with two non zero rows.

$\therefore \rho(A^*) = \rho(A) < 3$ (no. of unknowns)

$\therefore$ given system has infinite solns.

The system in reduced form is

$$2x+4y-z=9$$
$$7y-\frac{13}{2}z = -\frac{17}{2}$$

i.e. $14y-13z=-17$

Now two variables can be expressed in form of remaining one. Let $z=k$. Then $y=\frac{13k-17}{14}$

and $2x+\frac{2(13k-17)}{7}-k=9 \Rightarrow x = -\frac{19}{14}k+\frac{97}{14}$ Ans

Subst. arbitrary values to k, we get infinite solns.

(4) Determine the values of λ & μ s.t. the following system has (i) no solution (ii) a unique soln. (iii) infinite no. of soln.

$$2x - 5y + 2z = 8$$
$$2x + 4y + 6z = 5$$
$$x + 2y + \lambda z = \mu$$

Soln:

$$A^* = \begin{bmatrix} 2 & -5 & 2 & 8 \\ 2 & 4 & 6 & 5 \\ 1 & 2 & \lambda & \mu \end{bmatrix} \sim \begin{bmatrix} 2 & -5 & 2 & 8 \\ 0 & 9 & 4 & -3 \\ 0 & 9/2 & \lambda-1 & \mu-4 \end{bmatrix}$$

Alternatively, Can be solved by interchg 1st & 3rd row first

$$\sim \begin{bmatrix} 2 & -5 & 2 & 8 \\ 0 & 9 & 4 & -3 \\ 0 & 0 & \lambda-3 & \mu-\frac{5}{2} \end{bmatrix}$$

i) System has no soln. if

$\lambda - 3 = 0$ i.e. $\lambda = 3$ & $\mu \neq 5/2$

(ii) Unique soln. if $\lambda - 3 \neq 0$ i.e. $\lambda \neq 3$

(iii) Infinite soln. if $\lambda - 3 = 0$, $\mu - 5/2 = 0$

i.e. $\lambda = 3$ & $\mu = 5/2$.

(5) $x_1 + 2x_2 + x_3 = 2$, $3x_1 + x_2 - 2x_3 = 1$, $4x_1 - 3x_2 - x_3 = 3$
$2x_1 + 4x_2 + 2x_3 = 4$.

$$A^* = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 3 & 1 & -2 & 1 \\ 4 & -3 & -1 & 3 \\ 2 & 4 & 2 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & -5 & -5 & -5 \\ 0 & -11 & -5 & -5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & +1 & +1 & +1 \\ 0 & 0 & 6 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\rho(A) = 3$.

$$x_1 + 2x_2 + x_3 = 2$$
$$x_2 + x_3 = 1$$
$$x_3 = 1.$$

$\therefore x_2 = 0$, & $x_1 = 1$.

i.e. $x_1 = 1$, $x_2 = 0$, $x_3 = 1$ is the unique soln.

⑦ Determine the values of a and b for which the system

$$\begin{bmatrix} 3 & -2 & 1 \\ 5 & -8 & 9 \\ 2 & 1 & a \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b \\ 3 \\ -1 \end{bmatrix}$$

(i) has a unique soln. (ii) has no solution (iii) has infinitely many solutions

<u>Soln:</u> Aug. Matrix C $= \begin{bmatrix} 3 & -2 & 1 & b \\ 5 & -8 & 9 & 3 \\ 2 & 1 & a & -1 \end{bmatrix}$

$$\sim \begin{bmatrix} 3 & -2 & 1 & b \\ 2 & -6 & 8 & 3-b \\ 0 & 7 & a-8 & -1-3+b \end{bmatrix}$$

$$\sim \begin{bmatrix} 3 & -2 & 1 & b \\ 0 & 7 & a-8 & -1-3+b \\ 2 & -6 & 8 & 3-b \end{bmatrix}$$

$$\sim \begin{bmatrix} 3 & -2 & 1 & b \\ 0 & 7 & a-8 & -4+b \\ 0 & -\frac{14}{3} & \frac{22}{3} & \frac{9-5b}{3} \end{bmatrix} \quad R_3 \to R_3 - \frac{2}{3}R_1$$

$$\sim \begin{bmatrix} 3 & -2 & 1 & b \\ 0 & 7 & a-8 & -4+b \\ 0 & -14 & 22 & 9-5b \end{bmatrix}$$

$$\sim \begin{bmatrix} 3 & -2 & 1 & b \\ 0 & 7 & a-8 & -4+b \\ 0 & 0 & 6+2a & -3b+1 \end{bmatrix}$$

(i) The given system has a unique soln if Rank of A = Rank of C = 3.

i.e. $6+2a \neq 0 \Rightarrow a \neq -3$

(ii) No solution if Rank of A $\neq$ Rank of C

$6+2a=0$ & $-3b+1 \neq 0 \Rightarrow a=-3,\ b \neq +\frac{1}{3}$

(iii) Infinite solutions if Rank of A = Rank of C < 3 (No. of variables)

This is possible only if

$$2a+6=0 \quad \& \quad -3b+1=0$$

$$\Rightarrow \quad a=-3 \quad \text{and} \quad b=\frac{1}{3}.$$

System of Linear Algebraic Equations.

1. Gauss elimination method : In this method unknowns are eliminated successively and the system is reduced to an upper triangular system from which the unknowns are found by back substitution.

Ex. Apply Gauss elimination method to solve the equations

$x + 4y - z = -5$; $x+y-6z = -12$; $3x - y - z = 4$.

Soln: Given $x + 4y - z = -5$ —— (i)

$x + y - 6z = -12$ —— (ii)

$3x - y - z = 4$ —— (iii)

Step I. To eliminate x, operate (ii) − (i) and (iii) − 3 (i):

$-3y - 5z = -7$ —— (iv)

$-13y + 2z = 19$ (v)

Step II. To eliminate y, operate (v) − $\frac{13}{3}$ (iv):

$\frac{71}{3} z = \frac{148}{3}$ —— (vi)

Step III. By back substitution, we get

From (vi), $z = \frac{148}{71}$

From (iv), $y = -\frac{81}{71}$

From (i) $x = \frac{117}{71}$.

2. Gauss - Jordan method :

This is a modification of the Gauss elimination method. In this method, elimination of unknowns is performed not in the equation below but in the equations above also, ultimately reducing the system to a diagonal matrix form i.e. each equation involving only one unknown. From these equations, the unknowns x, y, z can be obtained readily.

Ex. Apply Gauss - Jordan method to solve the equations $x+y+z=9$; $2x-3y+4z=13$; $3x+4y+5z=40$.

Soln:

$$x+y+z=9 \quad \text{——— (i)}$$
$$2x-3y+4z=13 \quad \text{——— (ii)}$$
$$3x+4y+5z=40 \quad \text{——— (iii)}$$

Step I. To eliminate x from (ii) & (iii), operate (ii) $-$ 2(i) & (iii) $-$ 3(i). We get

$$x+y+z=9 \quad \text{— (iv)}$$
$$-5y+2z=-5 \quad \text{— (v)}$$
$$y+2z=13 \quad \text{— (vi)}$$

Step II. To eliminate y from (iv) & (vi), operate (iv) $+\frac{1}{5}$(v) and (vi) $+\frac{1}{5}$(v) :

$$x+\frac{7}{5}z=8 \quad \text{-- (vii)}$$
$$-5y+2z=-5 \quad \text{-- (viii)}$$
$$\frac{12}{5}z=12 \quad \text{-- (ix)}$$

Step III. To eliminate z from (vii) & (viii), operate (vii) $-\frac{7}{12}$(ix) and (viii) $-\frac{5}{6}$(ix) :

$$x=1,$$
$$-5y=\cancel{-15}$$
$$\frac{12}{5}z=12$$

Hence, soln. is $\boxed{x=1, \cancel{y}=3, z=5}$

Triangular Matrix Factorization Methods

This method is also known as the LU decomposition method or the method of factorization.

In this method, the coefficient matrix A of the system of equations $AX = B$ is decomposed into the product of a lower triangular matrix L and an upper triangular matrix U so that

$$A = LU \quad \text{---} (1)$$

where

$$L = \begin{bmatrix} l_{11} & 0 & 0 & \cdots & 0 \\ l_{21} & l_{22} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ l_{n1} & l_{n2} & \cdots & & 0 \end{bmatrix} \text{ and } U = \begin{bmatrix} u_{11} & u_{12} & u_{13} & \cdots & u_{1n} \\ 0 & u_{22} & u_{23} & \cdots & u_{2n} \\ 0 & 0 & u_{33} & \cdots & u_{3n} \\ \vdots & & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & u_{nn} \end{bmatrix}$$

Using the matrix multiplication and comparing corresponding elements in (1), we get the values of l_{ij} & u_{ij}.

To produce a unique soln, it is convenient to choose either $u_{ii} = 1$ or $l_{ii} = 1$; $1 \le i \le n$.

i) When we choose $l_{ii} = 1$, the method is called the Doolittle method,

ii) When we choose $u_{ii} = 1$, the method is called the Crout's method.

The given system of equations is

$$AX = B \Rightarrow LUX = B \quad \text{---} (3)$$

Let $UX = Y$, then eqn (3) becomes $LY = B$. --- (4)

The unknowns $y_1, y_2, \ldots y_n$ in (4) are determined by forward substitution and the unknowns $x_1, x_2, \ldots x_n$ in $UX = Y$ are obtained by back substitution.

Note – The method fails if any of l_{ii} or u_{ii} is zero.

Ex 1. Solve the following system of equations by Doolittle's method:

$$2x + 3y + z = 9$$
$$x + 2y + 3z = 6$$
$$3x + y + 2z = 8$$

Soln: We choose $l_{ii} = 1$ and write

$$\begin{bmatrix} 2 & 3 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$= \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

Equating, we get

$u_{11} = 2$, $u_{12} = 3$, $u_{13} = 1$

$l_{21}u_{11} = 1 \Rightarrow l_{21} = 1/2$

$l_{31}u_{11} = 3 \Rightarrow l_{31} = 3/2$

$l_{21}u_{12} + u_{22} = 2 \Rightarrow u_{22} = 1/2$

$l_{21}u_{13} + u_{23} = 3 \Rightarrow u_{23} = 5/2$

$l_{31}u_{12} + l_{32}u_{22} = 1 \Rightarrow l_{32} = -7$

$l_{31}u_{13} + l_{32}u_{23} + u_{33} = 2 \Rightarrow u_{33} = 18$

Thus, we get

$$A = LU = \begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 3/2 & -7 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 1 \\ 0 & 1/2 & 5/2 \\ 0 & 0 & 18 \end{bmatrix}$$

Given system is

$$AX = B \Rightarrow LUX = B \quad \text{—— (1)}$$

Let $UX = Y$ so that (1) becomes

$$LY = B$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 3/2 & -7 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 9 \\ 6 \\ 8 \end{bmatrix}$$

which gives $\underline{y_1 = 9}$

$$\frac{1}{2}y_1 + y_2 = 6 \Rightarrow y_2 = 6 - \frac{9}{2} = \underline{\frac{3}{2}}$$

$$\frac{3}{2}y_1 - 7y_2 + y_3 = 8 \Rightarrow y_3 = 8 - \frac{3}{2}.9 + 7.\frac{3}{2} = \frac{10}{2} = \underline{5}$$

Now $UX = Y$

$$\begin{bmatrix} 2 & 3 & 1 \\ 0 & 1/2 & 5/2 \\ 0 & 0 & 18 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 3/2 \\ 5 \end{bmatrix}$$

which gives $2x + 3y + z = 9$

$$\frac{1}{2}y + \frac{5}{2}z = 3/2$$

$$18z = 5$$

By back substitution,

$$x = \frac{35}{18}, \quad y = \frac{29}{18}, \quad z = \frac{5}{18}$$

Ex. 2. Solve, by Crout's method, the following system of equations:

$$x + y + z = 3$$
$$2x - y + 3z = 16$$
$$3x + y - z = -3$$

Soln. We chose $u_{ii} = 1$ and write

$$A = LU$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 3 \\ 3 & 1 & -1 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} l_{11} & l_{11}u_{12} & l_{11}u_{13} \\ l_{21} & l_{21}u_{12} + l_{22} & l_{21}u_{13} + l_{22}u_{23} \\ l_{31} & l_{31}u_{12} + l_{32} & l_{31}u_{13} + l_{32}u_{23} + l_{33} \end{bmatrix}$$

Equating, we get

$$l_{11} = 1, \quad l_{21} = 2, \quad l_{31} = 3$$

$$l_{11}u_{12} = 1 \Rightarrow u_{12} = 1$$

$l_{11} u_{13} = 1 \Rightarrow u_{13} = 1$

$l_{21} u_{12} + l_{22} = -1 \Rightarrow l_{22} = -3$

$l_{31} u_{12} + l_{32} = 1 \Rightarrow l_{32} = -2$

$l_{31} u_{13} + l_{22} u_{23} = 3 \Rightarrow u_{23} = -1/3$

$l_{31} u_{13} + l_{22} u_{23} + l_{33} = -1 \Rightarrow l_{33} = -14/3$

Thus, we get

$$A = LU = \begin{bmatrix} 1 & 0 & 0 \\ 2 & -3 & 0 \\ 3 & -2 & -\frac{14}{3} \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1/3 \\ 0 & 0 & 1 \end{bmatrix}$$

The given system is

$$AX = B$$

$$\Rightarrow LUX = B \quad \text{——— (1)}$$

let $UX = Y$ so that (1) becomes

$$LY = B$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & -3 & 0 \\ 3 & -2 & -14/3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 16 \\ -3 \end{bmatrix}$$

which gives $\underline{y_1 = 3}$, $2y_1 - 3y_2 = 16 \Rightarrow \underline{y_2 = -10/3}$

$3y_1 - 2y_2 - \frac{14}{3} y_3 = -3 \Rightarrow \underline{y_3 = 4}$

Now $UX = Y \Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1/3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -10/3 \\ 4 \end{bmatrix}$

which gives
$$x + y + z = 3$$
$$y - \frac{1}{3} z = -\frac{10}{3}$$
$$z = 4$$

By back substitution

$$x = 1, \quad y = -2, \quad z = 4.$$

Cholesky Method :

This method is also called square-root method.

Consider a system of equations

$$AX = B \qquad \text{--- (1)}$$

If the coefficient matrix A is symmetric and positive definite then A can be decomposed as

$$A = LL^T \qquad \text{--- (2)}$$

where L is a lower triangular matrix.

A may also be decomposed as $A = UU^T$ where U is an upper triangular matrix.

From (1) & (2),

$$LL^T X = B$$

$$LY = B \qquad \text{--- (3)}$$

where $L^T X = Y \qquad (4)$

The values y_i can be obtained by forward substitution and the solution x_i are obtained by back substitution.

Ex. Solve the following system of equations

$$x_1 + 2x_2 + 3x_3 = 5$$
$$2x_1 + 8x_2 + 22x_3 = 6$$
$$3x_1 + 22x_2 + 82x_3 = -10$$

Soln: $A = LL^T$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 22 \\ 3 & 22 & 82 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix}$$

$$= \begin{bmatrix} l_{11}^2 & l_{11} l_{21} & l_{11} l_{31} \\ l_{21} l_{11} & l_{21}^2 + l_{22}^2 & l_{21} l_{31} + l_{22} l_{32} \\ l_{31} l_{11} & l_{31} l_{21} + l_{32} l_{22} & l_{31}^2 + l_{32}^2 + l_{33}^2 \end{bmatrix}$$

Equating the corresponding elements, we get

$$l_{11}^2 = 1 \quad \Rightarrow \quad l_{11} = 1$$
$$l_{11} l_{21} = 2 \quad \Rightarrow \quad l_{21} = 2$$

$$\cancel{l_{11} l_{21} = 2 \quad \Rightarrow \quad l_{21} = 2}$$

$$l_{11} l_{31} = 3 \quad \Rightarrow \quad l_{31} = 3$$

$$l_{21}^2 + l_{22}^2 = 8 \quad \Rightarrow \quad l_{22} = 2$$

$$l_{31} l_{21} + l_{32} l_{22} = 22 \quad \Rightarrow \quad l_{32} = 8$$

$$l_{31}^2 + l_{32}^2 + l_{33}^2 = 82 \quad \Rightarrow \quad l_{33} = 3$$

Hence, $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 8 & 3 \end{bmatrix}$

$$L L^T X = B$$

$$L Y = B \quad \text{(Taking } L^T X = Y \text{)}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 8 & 3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ -10 \end{bmatrix}$$

$\Rightarrow y_1 = 5, \; y_2 = -2, \; y_3 = -3$ (by forward substitution)

From $L^T X = Y$, we get

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 8 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ -3 \end{bmatrix}$$

$$x_1 + 2x_2 + 3x_3 = 5$$

$$2x_2 + 8x_3 = -2$$

$$3x_3 = -3$$

By back substitution,

$$x_3 = -1,$$

$$2x_2 - 8 = -2 \Rightarrow x_2 = 3$$

$$x_1 + 6 - 3 = 5 \Rightarrow x_1 = 2$$

Thus,

$$\boxed{x_1 = 2, \; x_2 = 3 \quad x_3 = -1.}$$

Eigen values & Eigen vectors

If A is any square matrix of order n, then $|A-\lambda I| = 0$, where I is unit matrix of order n, is called the ~~coefficient matrix~~ characteristic equation ~~of~~ of A.

On expansion char. eqn. takes the form

$$(-1)^n\lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + \cdots + a_n = 0$$

$$\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

$\lambda = 2, 2, 8$

where a's are constants. The roots of this eqn. are called the characteristic roots, latent roots or eigen values of the matrix.

* If corresponding to an eigen value λ we can find a nonzero vector X, where $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ s.t.

$$[A-\lambda I]X = 0$$

then X is called the eigen vector corresponding the eigen value λ.

Note: 1 Corresponding to n distinct eigen values, we get n independent eigen vectors. But when two or more eigen values are equal, it may or may not be possible to get lin. ind. eigen vectors corresponding to the repeated roots.

2. If X_i is a soln. for an eigen value λ_i then it follows that cX_i is also a soln, where c is an arbitrary constant.

* Some Properties of eigen values:

(1) Any square matrix A and its transpose A' have the same eigen values

[The sum of the elements on the principal diagonal of a matrix is called the <u>trace of the matrix</u>

(2) The sum of the eigen values of a matrix is equal to the trace of the matrix.

(3) The product of the eigenvalues of a matrix A is equal to the determinant of A

(4) If $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the eigen values of A, then the eigen values of

(i) kA are $k\lambda_1, k\lambda_2, \ldots, k\lambda_n$

(ii) A^m are $\lambda_1^m, \lambda_2^m, \ldots, \lambda_n^m$

(iii) A^{-1} are $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \ldots, \frac{1}{\lambda_n}$.

(5) Eigen values of a triangular matrix are just the diagonal elements of the matrix.

<u>Ex. 1</u> $A = \begin{bmatrix} 2 & -3 & 1 \\ 3 & 1 & 3 \\ -5 & 2 & -4 \end{bmatrix}$ Eigen value. Ans. 0, 1, -2

(III) $A = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$ 1, 4

Ex. 2. $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$ Eigen values 2, 3, 5.

Ex 3. $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ 0, 3, 15

Ex 4. $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 3 & 2 & 3 \end{bmatrix}$ 0, 1, 5

* Show that the matrix $\begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$ has less than three lin. indep. eigen vectors. Also find them.

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Soln: $|A-\lambda I| = \begin{vmatrix} 3-\lambda & 10 & 5 \\ -2 & -3-\lambda & -4 \\ 3 & 5 & 7-\lambda \end{vmatrix} = 0$

$$\Rightarrow -\lambda^3 + 7\lambda^2 - 16\lambda + 12 = 0$$

$$\Rightarrow \lambda^3 - 7\lambda^2 + 16\lambda - 12 = 0$$

$$\Rightarrow \lambda^3 - 2\lambda^2 - 5\lambda^2 + 10\lambda + 6\lambda - 12 = 0$$

$$\Rightarrow \lambda^2(\lambda-2) - 5\lambda(\lambda-2) + 6(\lambda-2) = 0$$

$$\Rightarrow (\lambda-2)(\lambda^2 - 5\lambda + 6) = 0$$

$$\Rightarrow \lambda = 2, 2, 3.$$

Eigen vectors are obtained from $[A-\lambda I]X = 0$

i.e. $\begin{bmatrix} 3-\lambda & 10 & 5 \\ -2 & -3-\lambda & -4 \\ 3 & 5 & 7-\lambda \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

i) Corresponding to $\lambda = 2$,

$$\begin{bmatrix} 1 & 10 & 5 \\ -2 & -5 & -4 \\ 3 & 5 & 5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Coeff. matrix $\begin{bmatrix} 1 & 10 & 5 \\ -2 & -5 & -4 \\ 3 & 5 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 10 & 5 \\ 0 & 15 & 6 \\ 0 & -25 & -10 \end{bmatrix} \sim \begin{bmatrix} 1 & 10 & 5 \\ 0 & 5 & 2 \\ 0 & -5 & -2 \end{bmatrix}$

$$\sim \begin{bmatrix} 1 & 10 & 5 \\ 0 & 5 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

which is in echelon form. Rank of coeff. matrix $= 2$.

$\therefore$ no. of indep. eigen values is 1.

$$\begin{bmatrix} 1 & 10 & 5 \\ 0 & 5 & 2 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

gives, $x_1 + 10x_2 + 5x_3 = 0$

$5x_2 + 2x_3 = 0 \Rightarrow x_2 = -\frac{2}{5}x_3$

& $x_1 = -10x_2 - 5x_3 = -10\left(-\frac{2}{5}x_3\right) - 5x_3$

$= \cancel{\frac{4}{5}x_3} - x_3$

choosing $x_3 = k_1$

$$x_1 = \cancel{\tfrac{5}{3}k_1}\ -k_1, \quad x_2 = -\tfrac{2}{5}k_1, \quad x_3 = k_1$$

$\therefore$ Eigen vector corresponding to $\lambda = 2$ is

$$X_1 = \begin{bmatrix} -k_1 \\ -\frac{2}{5}k_1 \\ k_1 \end{bmatrix} = k_1 \begin{bmatrix} -5 \\ -2 \\ 5 \end{bmatrix}$$

ii) For $\lambda = 3$,

$$\begin{bmatrix} 0 & 10 & 5 \\ -2 & -6 & -4 \\ 3 & 5 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 2 \\ 0 & 2 & 1 \\ 3 & 5 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 2 \\ 0 & 2 & 1 \\ 0 & -4 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 2 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

eqns in reduced form

$$x_1 + 3x_2 + 2x_3 = 0$$

$$2x_2 + x_3 = 0 \Rightarrow x_3 = -2x_2$$

$$x_1 = -3x_2 - 2x_3 = -3x_2 - 2(-2x_2) = x_2$$

Choosing $x_2 = k_2$

$$x_1 = k_2, \quad x_2 = k_2, \quad x_3 = -2k_2$$

$\therefore$ Eigen vector is $X_2 = k_2 \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$

Ans.

* Find the eigen vectors for the matrix $A = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$ UPTU 2014

* Find the eigen values & vectors of $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ UPTU -2015

Eigen Value Problems

Power method: In many engineering problems, it is required to compute the numerically largest eigen value and the corresponding eigen value.

Procedure – Start with a column vector X which is as near the solution as possible and evaluate AX which is written as $\lambda^{(1)} x^{(1)}$ after normalization. This gives the first approximation $\lambda^{(1)}$ to the eigen value and $x^{(1)}$ to the eigen vector. Similarly, evaluate $Ax^{(1)} = \lambda^{(2)} x^{(2)}$ which gives the second approximation. Repeat this process till $[x^{(r)} - x^{(r-1)}]$ becomes negligible. Then $\lambda^{(r)}$ will be the largest eigen value and $x^{(r)}$, the corresponding eigen vector.

This iterative procedure for finding the dominant eigen value of a matrix is known as Rayleigh's power method.

Ex. Determine the largest eigen value and the corresponding eigen vector of the matrix

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

Soln: Let the initial eigen vector be $[1, 0, 0]^T$.

Then
$$AX = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -0.5 \\ 0 \end{bmatrix}$$

Thus, first approximation is

$$\lambda^{(1)} = 2, \quad X^{(1)} = [1, -0.5, 0]^T$$

Hence, $Ax^{(1)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -0.5 \\ 0 \end{bmatrix} = \begin{bmatrix} 2.5 \\ -2 \\ 0.5 \end{bmatrix} = 2.5 \begin{bmatrix} 1 \\ -0.8 \\ 0.2 \end{bmatrix}$

Thus, second approximation

$\lambda^{(2)} = 2.5, \quad x^{(2)} = [1, -0.8, 0.2]^T$

$Ax^{(2)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -0.8 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 2.8 \\ -2.8 \\ 1.2 \end{bmatrix} = 2.8 \begin{bmatrix} 1 \\ -1 \\ 0.43 \end{bmatrix}$

Third approximation,

$\lambda^{(3)} = 2.8, \quad x^{(3)} = [1, -1, 0.43]$

$Ax^{(3)} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 0.43 \end{bmatrix} = \begin{bmatrix} 3 \\ -3.43 \\ 1.86 \end{bmatrix} = 3.43 \begin{bmatrix} .87 \\ -1 \\ .54 \end{bmatrix}$

Similarly,

$Ax^{(4)} = 3.41 \begin{bmatrix} 0.80 \\ -1 \\ 0.61 \end{bmatrix}, \quad Ax^{(5)} = 3.41 \begin{bmatrix} 0.76 \\ -1 \\ 0.65 \end{bmatrix}$

$Ax^{(6)} = 3.41 \begin{bmatrix} 0.74 \\ -1 \\ 0.67 \end{bmatrix}$

Hence, the largest eigen value is 3.41 and the corresponding eigen vector is $[0.74, -1, 0.67]^T$.

Ex. Find the largest eigen-value and the corresponding eigen-vector of the matrices

i) $\begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix}$ ii) $\begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

Ans: i) 11.66, $[0.025, 0.422, 1]^T$

ii) 4, $[2, 1, 0]^T$.

Spline Interpolation

In the interpolation methods discussed so far, a nth order polynomial passing through $(n+1)$ given data points was obtained. In certain cases, these polynomials tend to give erroneous results. Further it was found that a low order polynomial approximation in each subinterval provides a better approximation to the tabulated function than fitting a single high-order polynomial to the entire interval. Such an interpolation is called piecewise polynomial interpolation and the spline functions are such piecewise connecting polynomials.

The points at which two connecting splines meet are called knots. The connecting polynomials could be of any degree and therefore, we have different types of splines: linear, quadratic, cubic, quintic, etc. Of these, the cubic splines have been found to be most popular in engineering applications.

1. Linear Splines:

let the given data points be $(x_i, y_i), i=0,1,2,--n$.

where $a = x_0 < x_1 < --- x_n = b$, Then, the linear spline is given by

$$S_i(x) = y_{i-1} + \frac{y_i - y_{i-1}}{x_i - x_{i-1}}(x - x_{i-1})$$

~~betw~~ in the interval $[x_{i-1}, x_i]$

Note: $S_i(x)$ is continuous at both the end points.

2. Quadratic Splines:

Given data points $(x_i, y_i), i=0,1,2,--n$.

$$h_i = x_i - x_{i-1}, \quad i=1,2,---,n.$$

Then, the quadratic polynomial is

$$S_i(x) = \frac{1}{h_i}\left[-\frac{(x_i - x)^2}{2} m_{i-1} + \frac{(x - x_{i-1})^2}{2} m_i\right] + y_{i-1} + \frac{h_i}{2} m_{i-1}$$

where $$m_{i-1} + m_i = \frac{2}{h_i}(y_i - y_{i-1}), \quad i=1,2,--,n.$$

and $$m_0 = m_1.$$

3. Cubic Splines:

Given data points (x_i, y_i) and let $S_i(x)$ be the cubic spline defined in the interval $[x_{i-1}, x_i]$. The conditions for the natural cubic spline are

i) $S_i(x)$ is almost a cubic in each subinterval $[x_{i-1}, x_i]$

ii) $S_i(x_i) = y_i, \quad i=0,1,2,--n.$

iii) $S_i(x), S_i'(x)$ and $S_i''(x)$ are continuous in $[x_0, x_n]$

iv) $S_i''(x_0) = S_i''(x_n) = 0.$

The interpolating cubic polynomial is given by

$$S_{i+1}(x) = \frac{(x_{i+1} - x)^3}{6h} M_i + \frac{(x - x_i)^3}{6h} M_{i+1}$$
$$+ \frac{x_{i+1} - x}{h}\left(y_i - \frac{h^2}{6} M_i\right) + \frac{x - x_i}{h}\left(y_{i+1} - \frac{h^2}{6} M_{i+1}\right)$$

where $M_{i-1} + 4M_i + M_{i+1} = \frac{6}{h^2}\left(y_{i-1} - 2y_i + y_{i+1}\right)$

$i = 1, 2, ---, n-1$

When $M_0 = M_n = 0$, the cubic spline is called as a natural cubic spline. Also, $M_0 = f''(x_0)$, $M_n = f''(x_n)$. These are called end conditions.

Ex. 1. From the following data:

$x:$ 1 2 3

$y:$ -8 -1 18

find i) the linear spline

ii) the quadratic spline

and iii) the natural cubic spline

Also, determine the approximate values of $y(2.5)$ and $y'(0)$.

Soln: i) Linear spline

We know that $S_i(x) = y_{i-1} + \frac{y_i - y_{i-1}}{x_i - x_{i-1}}(x - x_{i-1})$

in the interval $[x_{i-1}, x_i]$ $i = 1, 2, -- n$

(a) In the interval $[1, 2]$

$S_1(x) = y_0 + \frac{y_1 - y_0}{x_1 - x_0}(x - x_0) = -8 + \frac{-1-(-8)}{2-1}(x-1)$

$= -8 + 7(x-1) = 7x - 15.$

i.e. $S_1(x) = 7x - 15$

(b) In the interval $[2, 3]$

$S_2(x) = y_1 + \frac{y_2 - y_1}{y_2 - y_1}(x - x_1) = -1 + \frac{18-(-1)}{3-2}(x-2)$

$= -1 + 19(x-2) = 19x - 39$

i.e. $S_2(x) = 19x - 39.$

Since, $x = 2.5$ is in interval $[2, 3]$, we have

$y(2.5) = S_2(2.5) = 19(2.5) - 39 = 8.5$

Also, $y'(2.0) = 19.$

ii) Quadratic spline

$x_0 = 1, \quad x_1 = 2, \quad x_2 = 3$

$y_0 = -8, \quad y_1 = -1, \quad y_2 = 18$

$h = 1$ as data is equispaced

$$S_i(x) = \frac{1}{h}\left[-\frac{(x_i - x)^2}{2} m_{i-1} + \frac{(x - x_{i-1})^2}{2} m_i\right] + y_{i-1} + \frac{h}{2} m_{i-1}$$

& $m_{i-1} + m_i = \frac{2}{h}(y_i - y_{i-1})$; $i = 1, 2, \ldots$ and $m_0 = m_1$

Calculate m_0, m_1, m_2 first.

For $i=1$, $m_0 + m_1 = 2(y_1 - y_0) = 2((-1) - (-8)) = 14$ —(i)

for $i=2$, $m_1 + m_2 = 2(y_2 - y_1) = 2(18 - (-1)) = 38$ —(ii)

from (i) $m_0 = m_1 = 7$.

from (ii) $m_2 = 38 - 7 = 31$

$$\therefore S_1(x) = \frac{1}{1}\left[-\frac{(x_1 - x)^2}{2} m_0 + \frac{(x - x_0)^2}{2} m_1\right] + y_0 + \frac{h}{2} m_0$$

$$= \left[-\frac{(2-x)^2}{2} \cdot 7 + \frac{(x-1)^2}{2} 7\right] + (-8) + \frac{1}{2} \times 7$$

$$= \frac{7}{2}\left[-4 + 4x - \cancel{x^2} + \cancel{x^2} - 2x + 1\right] - \frac{9}{2}$$

$$= \frac{7}{2}[2x - 3] - \frac{9}{2} = \frac{1}{2}[14x - 21 - 9]$$

$$= \frac{14x - 30}{2} = 7x - 15.$$

$$S_2(x) = \frac{1}{1}\left[-\frac{(x_2 - x)^2}{2} m_1 + \frac{(x - x_1)^2}{2} m_2\right] + y_1 + \frac{1}{2} m_1$$

$$= \left[-\frac{(3-x)^2}{2} \cdot 7 + \frac{(x-2)^2}{2} \cdot 31\right] + (-1) + \frac{7}{2}$$

$$= \frac{1}{2}\left[(-9 + 6x - x^2)7 + (x^2 - 4x + 4)31\right] + \frac{5}{2}$$

$$= \frac{1}{2}\left[24x^2 - 82x + 61\right] + \frac{5}{2}$$

$$= 12x^2 - 41x + 33$$

Thus, quadratic spline is

$$S(x) = \begin{cases} 7x - 15, & 1 \le x \le 2 \\ 12x^2 - 41x + 33, & 2 \le x \le 3 \end{cases}$$

Hence, $y(2.5) = S_2(2.5) = 12 \times 6.25 - 41 \times 2.5 + 33$

$= 75 - 102.5 + 33 = 5.5$.

Thus, $\boxed{y(2.5) = 5.5}$

$2 \in [2,3]$.

$y'(2) = S_2'(2) = 24x - 41 = 48 - 41 = 7.0$.

Thus, $\boxed{y'(2) = 7.0}$.

iii) Cubic spline :

$$S_{i+1}(x) = (x_{i+1} - x)^3 M_i + \frac{(x-x_i)^3}{6h} M_{i+1} + \frac{(x_{i+1}-x)}{h}\left(y_i - \frac{h^2}{6} M_i\right) + \frac{(x-x_i)}{h}\left(y_{i+1} - \frac{h^2}{6} M_{i+1}\right) \quad (1)$$

$i = 0, 1, 2, \ldots$

where $M_{i-1} + 4M_i + M_{i+1} = \frac{6}{h^2}\left(y_{i-1} - 2y_i + y_{i+1}\right)$ — $i = 1, 2, \ldots n-1$ (2)

& $M_0 = M_n = 0$

Here, $x_0 = 1$, $x_1 = 2$, $x_3 = 3$

$y_0 = -8$, $y_1 = -1$, $y_2 = 18$

$i = 0, 1, 2$.

~~(a) In interval [1, 2]~~

~~$S(x) = (x_1$~~

To find M_0, M_1, M_2 : $M_0 = M_2 = 0$

From (2), $i = 1$, gives

$M_0 + 4M_1 + M_2 = \frac{6}{1}(y_0 - 2y_1 + y_2) = 6(-8 - 2(-1) + 18)$
$= 72$

$\Rightarrow 0 + 4M_1 + 0 = 72 \Rightarrow M_1 = 4$,

(a) In interval $[1, 2]$

$$S_1(x) = (x_1 - x)^3 M_0 + \frac{(x-x_0)^3}{6} M_1 + \frac{x_1 - x}{h}\left(y_0 - \frac{1}{6}M_0\right) + \frac{(x-x_0)}{1}\left(y_1 - \frac{1}{6}M_1\right)$$

$= (2-x)^3 \cdot 0 + \frac{(x-1)^3}{6} \times 18 + (2-x)\left(-8 - \frac{1}{6} \cdot 0\right) + (x-1)\left(-1 - \frac{1}{6} \times 18\right)$

$= \frac{(x-1)^3}{6} \times 18 + 8(x-2) + (x-1)(-4)$

$\Rightarrow \boxed{S_1(x) = (x-1)^3 + 8(x-2) - 4(x-1)} = x^3 - 1 - 3x(x-1) + 4x - 12$

$= x^3 - 3x^2 + 7x - 13$

$\therefore \boxed{S_1(x) = x^3 - 3x^2 + 7x - 13}$

(b) In the interval $[2, 3]$

$$S_2(x) = \frac{(x_2-x)^3}{6} M_1 + \frac{(x-x_1)^3}{6} M_2 + (x_2 - x)\left(y_1 - \frac{1}{6}M_1\right) + \frac{(x-x_1)}{1}\left(y_2 - \frac{1}{6}M_2\right)$$

$$= \frac{(3-x)^3}{6} \times 18 + \frac{(x-2)^3}{6} \times 0 + (3-x)\left(-1 - \frac{1}{6} \times 18\right) + (x-2)\left(18 - \frac{1}{6}\times 0\right)$$

$$= 3(3-x)^3 + (3-x)(-4) + (x-2) \times 18$$

$$= 3(3-x)^3 + 4(x-3) + 18(x-2) = 3(3-x)^3 + 22x - 48$$

$$\therefore S_2(x) = 3(3-x)^3 + 22x - 48 = 3\left[27 - x^3 - 9x(3-x)\right] + 22x - 48$$

$$= 3\left[27 - x^3 - 27x + 9x^2\right] + 22x - 48$$

$$= -3x^3 + 27x^2 - 59x + 33$$

$$\therefore \boxed{S_2(x) = -3x^3 + 27x^2 - 59x + 33}$$

Thus, the cubic spline is

$$S(x) = \begin{cases} x^3 - 3x^2 + 7x - 13, & 1 \le x \le 2 \\ -3x^3 + 27x^2 - 59x + 33, & 2 \le x \le 3. \end{cases}$$

$\because 2.5 \in [2, 3]$. Hence

$$y(2.5) = S_2(2.5) = -3(2.5)^3 + 27(2.5)^2 - 59 \times 2.5 + 33 = 7.375$$

$$= \cancel{-46.875 + 56.25 - 135 + 33}$$

$$\cancel{= 4.125}$$

Thus, $\boxed{y(2.5) = 7.375}$

$$y'(2) = S_2'(2) = -9(2)^2 + 54 \times 2 - 59 = 13$$

Thus, $\boxed{y'(2) = 13}$

Note that the tabulated fn is $y = x^3 - 9$ and hence the exact values of $y(2.5)$ & $y'(2)$ are 6.625 & 12.0 respectively. So, errors are 0.75 & 1.0 respectively.

Ex.2. Obtain the cubic spline for the data given below

x:	0	1	2	3
$f(x)$:	1	4	10	8

under the end condition $f''(0) = 0 = f''(3)$. Also, estimate $f(1.5)$

Ex 1. The following values of x & $f(x)$ are given:

x:	1	2	3	4
$f(x)$:	1	2	5	11

Given that $f''(1) = 0 = f''(4)$. Find the cubic polynomial and evaluate $y(1.5)$ & $y'(3)$.

Soln; Here $h=1$, $n=3$, we obtain from above (2)

$$M_{i-1} + 4M_i + M_{i+1} = 6\left(f(x_{i-1}) - 2f(x_i) + f(x_{i+1})\right)$$

$i = 1, 2$

For $i=1$, $M_0 + 4M_1 + M_2 = 6\left(f(x_0) - 2f(x_1) + f(x_2)\right)$

$= 6(1 - 4 + 5) = 12$

i.e. $4M_1 + M_2 = 12$ —— (α)

For $i=2$, $M_1 + 4M_2 + M_3 = 6\left(f(x_1) - 2f(x_2) + f(x_3)\right)$

$= 6(2 - 10 + 11) = 18$

i.e. $M_1 + 4M_2 = 18$ —— (β)

From (α) & (β) $M_1 = 2$, $M_2 = 4$ As $M_0 = M_3 = 0$.

On interval [1, 2]

$$F(x) = \frac{(2-x)^3}{6} M_0 + \frac{(x-1)^3}{6} M_1 + (2-x)\left(f(x_0) - \frac{1}{6}M_0\right) + (x-1)\left(f(x_1) - \frac{1}{6}M_1\right)$$

$$= 0 + \frac{x^3 - 3x^2 + 3x - 1}{3} + (2-x)\left(1 - \frac{1}{6}\times 0\right) + (x-1)\left(2 - \frac{1}{3}\right)$$

$$= \frac{1}{3}\left[x^3 - 3x^2 + 3x - 1 + 6 - 3x + 5x - 5\right]$$

$$= \frac{1}{3}\left[x^3 - 3x^2 + 5x\right]$$

On interval [2, 3]

$$F(x) = \frac{(3-x)^3}{6} M_1 + \frac{(x-2)^3}{6} M_2 + (3-x)\left(f(x_1) - \frac{1}{6}M_1\right) + (x-2)\left(f(x_2) - \frac{1}{6}M_2\right)$$

$$= \frac{1}{3}\left[(3-x)^3 + \frac{2}{3}(x-2)^3 + (3-x)\left(2 - \frac{1}{3}\right) + (x-2)\left(5 - \frac{2}{3}\right)\right]$$

$$= \frac{1}{3}\left[27 - 27x + 9x^2 - x^3 + 2x^3 - 16 - 12x^2 + 24x + 15 - 5x + 13x - 26\right]$$

$$= \frac{1}{3}\left[x^3 - 3x^2 + 5x\right]$$

On interval (3, 4)

$$F(x) = \frac{(4-x)^3}{6} M_2 + \frac{(x-3)^3}{6} M_3 + (4-x)\left(5 - \frac{1}{6}M_2\right) + (x-3)\left(11 - \frac{1}{6}M_3\right)$$

$$= \frac{2}{3}\left(64 - x^3 - 48x + 12x^2\right) + \frac{1}{3}\left(52 - 13x\right) + 33x - 99$$

$$= \frac{1}{3}\left(-2x^3 + 24x^2 - 76x + 81\right).$$

Thus cubic polynomial is

$$F(x) = \begin{cases} \frac{1}{3}\left(x^3 - 3x^2 + 5x\right) & 1 \le x \le 2 \\ \frac{1}{3}\left(x^3 - 3x^2 + 5x\right) & 2 \le x \le 3 \\ \frac{1}{3}\left(-2x^3 + 24x^2 - 76x + 81\right) & 3 \le x \le 4 \end{cases}$$